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Roll No

EC-703(C)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Probability Theory and Stochastic Processor

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) State and prove Bayes's Theorem.
b) A missile can be accidentally launched if two relays A and B both have failed. The probabilities of A and B failing are known to be 0.01 and 0.03 respectively. It is also known that B is more likely to fail (probability 0.06) if A failed.
 - i) What is the probability of an accidental missile launch?
 - ii) What is the probability that A will fail if B has failed?
 - iii) Are events "A fails" and "B fails" statistically independent?
2. a) Find the movement generating function of a uniform distribution and hence find its mean obtain the mean.
b) Obtain mean value of a sum of N weighted random variables. Also define joint moments about the origin.
3. a) Explain Chebyshev's Inequality.
b) What are the axioms of probability? Give engineering examples.

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4.
 - a) List the properties of N random variable.
 - b) In a control system, a random voltage X is known to have a mean value of $-2V$ and a second moment of $9V^2$. If the voltage X is amplified by an amplifier that gives an output $Y = 1.5X + 2$, find the variance of Y.
5.
 - a) Explain the concept and classification of stochastic process.
 - b) Derive the relation between cross power density spectrum and cross correlation function of a random process.
6.
 - a) Define in-phase component and quadrature-phase component.
 - b) Write short notes on the following:
 - i) Relationship between power spectrum and autocorrelation function.
 - ii) Poisson random process.
7.
 - a) Explain the following terms:
 - i) Variance
 - ii) Skew
 - b) Explain the properties of joint distribution.
8.
 - a) List the properties of autocorrelation function.
 - b) Explain the Rayleigh probability density function.
